

Product Requirements Document (PRD)

Project Name: SkillSwap

Status: MVP Implemented

Target Platform: Web Application (Desktop & Mobile Responsive)

Tech Stack: MERN Stack (MongoDB, Express.js, React.js, Node.js)

1. Executive Summary & Vision

Most skill-sharing currently relies on informal personal networks. Existing digital platforms lack proper matching, trust, and tracking mechanisms, making skill exchanges unreliable and difficult to manage. **SkillSwap** is a web-based platform designed to democratize learning by allowing individuals to trade skills they possess for skills they need, entirely without monetary transactions.

2. Product Strategy & Trade-offs

- **The Trust Problem:** Free platforms often suffer from ghosting and poor quality. To solve this, the MVP prioritizes a rigid **Review and Rating system** to build community trust.
- **The Gamification Loop:** To incentivize users to actually teach rather than just consume, we built a **Community Leaderboard** where top contributors earn points by successfully completing skill swaps.
- **Future AI Integration (Trade-off):** To ship the MVP faster, we used a deterministic matching logic system. However, the data architecture was specifically designed to support AI-powered recommendations and chatbot assistance in the next roadmap phase.

3. Core Functional Requirements (FRs)

Module 1: Dynamic Profiles & Discovery

- **FR-1.1:** Users must create a profile explicitly listing "Skills I Can Teach" and "Skills I Want to Learn".
- **FR-1.2:** A centralized "Market" dashboard allowing users to search for specific skills or people.
- **FR-1.3:** A matching mechanism that connects users based on their offered and required skills.

Module 2: The Swap Workflow & Communication

- **FR-2.1:** Users can send a "Propose a Swap" request to initiate a skill exchange.
- **FR-2.2:** An "Inbox" UI to track and manage incoming and sent swap requests.
- **FR-2.3:** A secure, in-app chat module that only unlocks *after* a skill swap request is formally accepted by both parties.

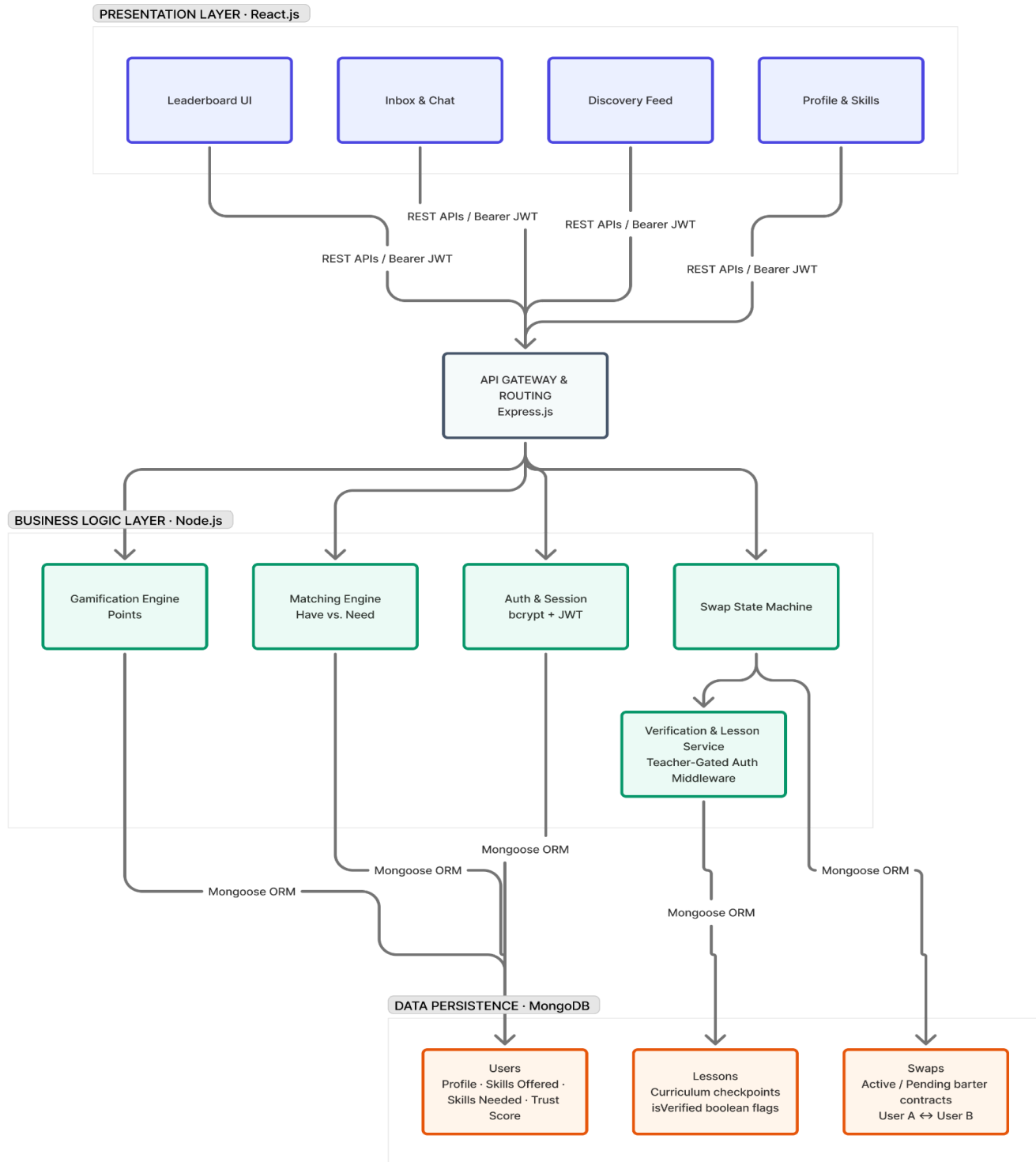
Module 3: Accountability & Gamification

- **FR-3.1:** A structured lesson tracking system to monitor user progress for online lectures.
- **FR-3.2:** Instructor-gated verification (The teaching user must verify the learner understood the topic before it is marked complete).
- **FR-3.3:** Leaderboard point distribution triggered only upon successful, verified swap completions.

4. System Architecture

The application utilizes a secure client-server model:

1. **Frontend (User Panel):** React.js interface handling skill listings, swap requests, and chat.
2. **Backend Server:** Express/Node.js handling matching logic, user management, and reviews.
3. **Database (MongoDB):** Storing user profiles, swap requests, chat messages, and skills.
4. **Admin Panel:** Infrastructure to manage users and handle content disputes.



5. Future Scope & Roadmap

To scale engagement and platform utility, the next phases will introduce:

1. **AI-Powered Matching:** Upgrading the current search logic to an AI recommendation

engine.

2. **Advanced Analytics & Notifications:** Real-time push notifications and user learning analytics.
3. **Mobile Expansion:** Developing native mobile support for on-the-go learning.